General Support, Timber Support

26/08/2019

Requirement of Support I

Supports depends on followings

*Depth

*Local geological srructure (tectonics)

*Weathering

*Presence of water

*Physical & Mechanical properties of rock.

*Bedding condition.

*Dimension of excavation

*Method of excavation.

Design of Support

*Direction & magnitude of rock pressure.

*Strength of rock around road ways & tunnel.

*Life span of opening.

*Joint plane / bedding plan

*Stress state of rock mass.

*Mechanical properties of rock.

*Shape, dimension & location of excavation.

*Support work as reinforcement (strengthening rock & control bulking).

*Holding & retaining rock mass.

Design Of Support I

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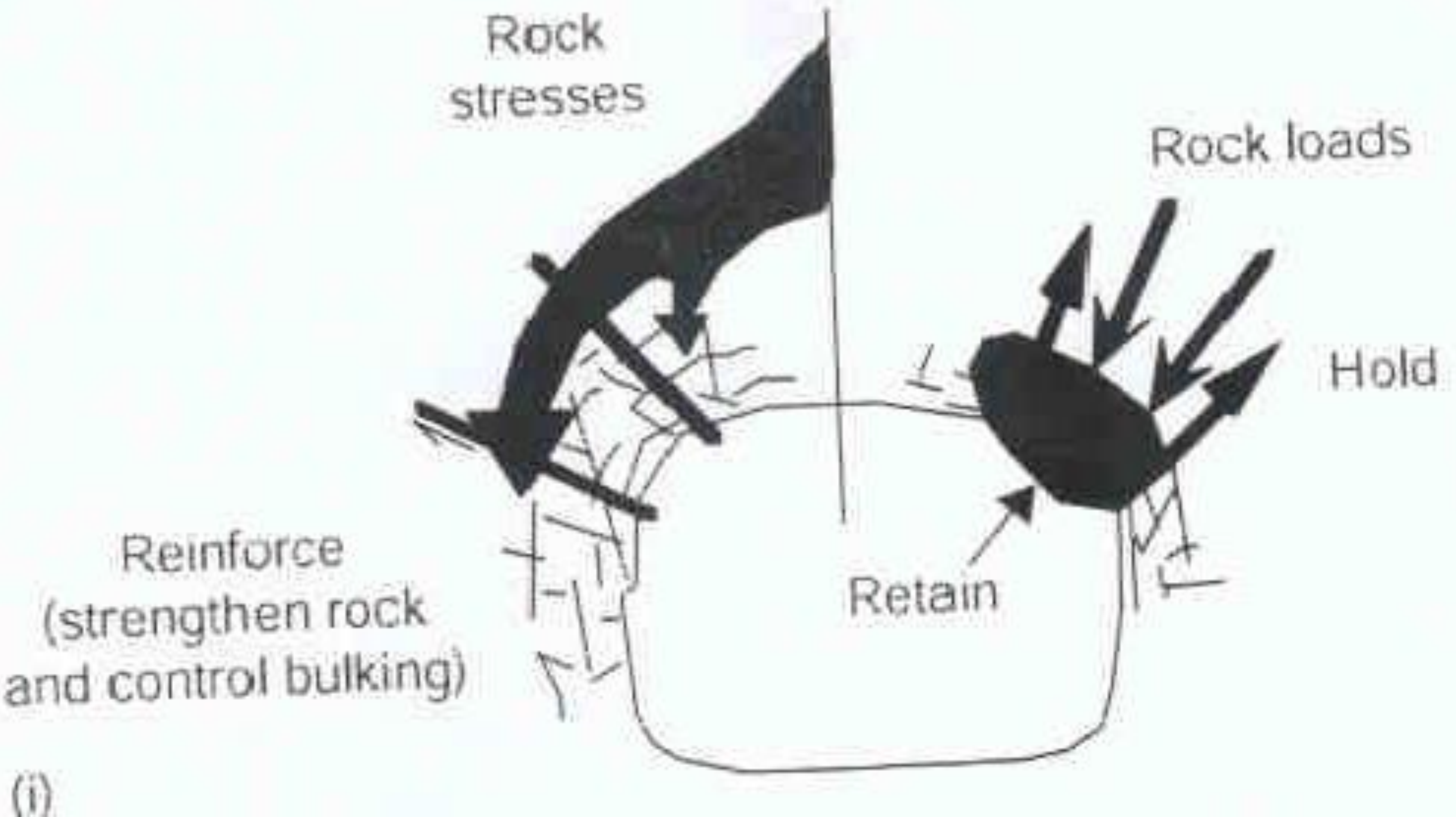

Support Design I

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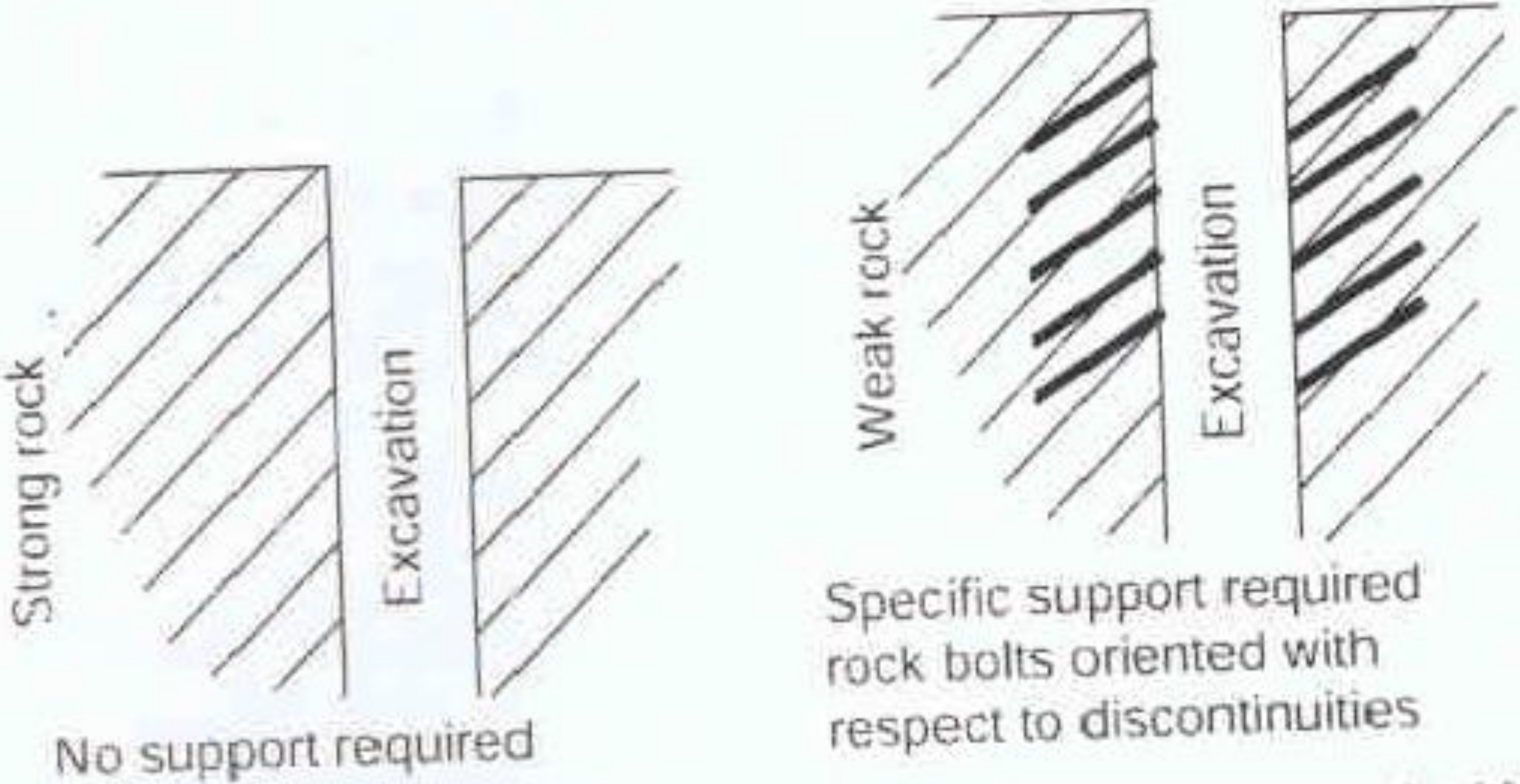

(ii) Single set of discontinuities in strong rock

(iii) Single set of discontinuities in weak rock

Importance of Support

*Support determine safety of work.

*Support effect production cost.

*Support reduces dilution & losses.

*Support effects intensity of mining & productivity of mines.

*Element of rock pressure (stress field) in surrounding rock, pillars & artificial support constitute a support net work in mines.

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Self Support By In Situ Rock

*Excavation design should utilize, in situ rock as much as possible as support.

*Excavation method should not set loose rock & reduce its strength.

*Proper excavation shape to be selected, so that excavation should be self supported as much as practical.

*Blasting method should be so selected, that back & side wall does not get effected / deteriorate.

*Size of excavation should not be unnecessarily large, demanding artificial support.

Support by Natural Pillars I

*Microwave heating can fragment hard rock.

*It can also weaken rock to improve performance of other mining method or secondary breaking.

*Micro wave can penetrate rock.

*Electro magnetic energy is converted by rapid vibration of molecules at a location (hot Spot inside the rock.

*Hot rock then expands according to its specific thermal expansion.

*Surrounding cool rock does not expand.

Rock (Natural Pillar) I

*Pillars are left in mine workings are made of in situ or waste rock.

*Thesis pillars are rigid & does not yield.

*Board & Pillar, Room & Pillar, Post Pillar are mining method with rock pillars to maintain stability of stope.

*Ore pillars may be left for ever or can be recovered after wards.

*Protective pillars are left to protect shaft (shaft pillar) or some other important mine structure in underground

Rock (Natural Pillar) II

*Level pillars are left to protect main levels, driver, cross cut etc.

*Crown pillars are left to protect upper level of stope & top part of stope.

*Sill pillars are left to protect lower level & bottom part of stope.

*Rib pillar / Block Pillar / Side Pillar are left to protect two adjacent stope & blocks.

*Support with ore pillar is simple & easy to do.

*Ore pillar causes loss of ore & inefficient mining work.

Rock (Natural Pillar) III

*Support design should consider minimum ore loss in pillar.

*Mining method should allow recovery of ore pillars at end of working.

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Artificial Support I

*Artificial supports are mine support made from support material brought from surface or other part of mines.

*Artificial supports are required to maintain stability of mine development & stope.

*Artificial support plays very important role in systematic support plan for mine development & stoping.

*Role of artificial support is reduced in caving method.

*Support holds loose rock, ket boulders etc.

Artificial Support II

*Support reinforce rock mass & control bulking.

*Support retains broken or unstable rock between holding & reinforcement to form stratified arch.

*Artificial supports are classified by material of construction.

*Life of support.

*Rigid or yielding type.

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Brick & Stone Support I

*Brick or dressed 7 sized stone can be used for support.

*These materials are used to support / lines shaft, drive by arching.

*Arching is predominantly uded at mine portal.

*Transportation & use of these material for support are labour intensive.

*Low cost skilled labour are required to use thses material for support.

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Timber Support I

*Locally available wood is used for timber support.

*“Sal” is widely used, especially at eastern part of India.

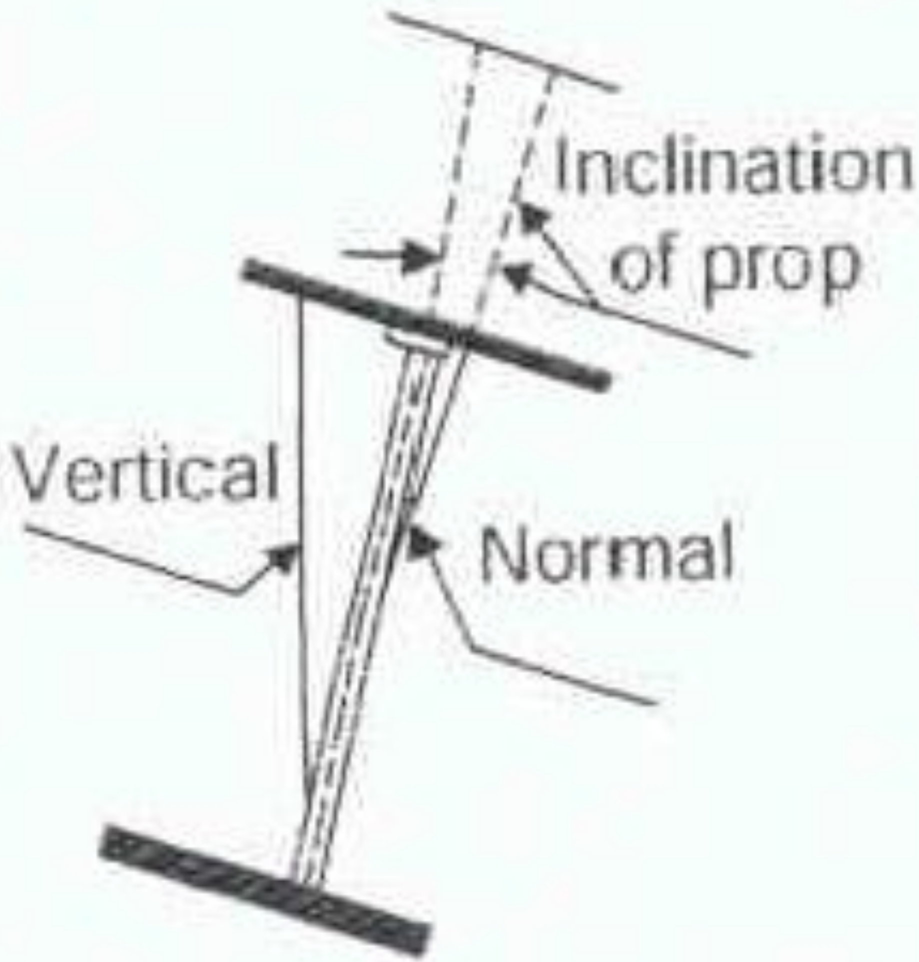
*Sal timber has very tough fiber, generally straight tree & naturally persistent to fungi etc.

*Timber strength depends on fiber structure.

*Presence of knot, non-concentric layers, fibers inclination, out side & inside cracks are common defects.

*These defects further weakens timber.

Wooden Support I



(a) Wooden prop & its setting

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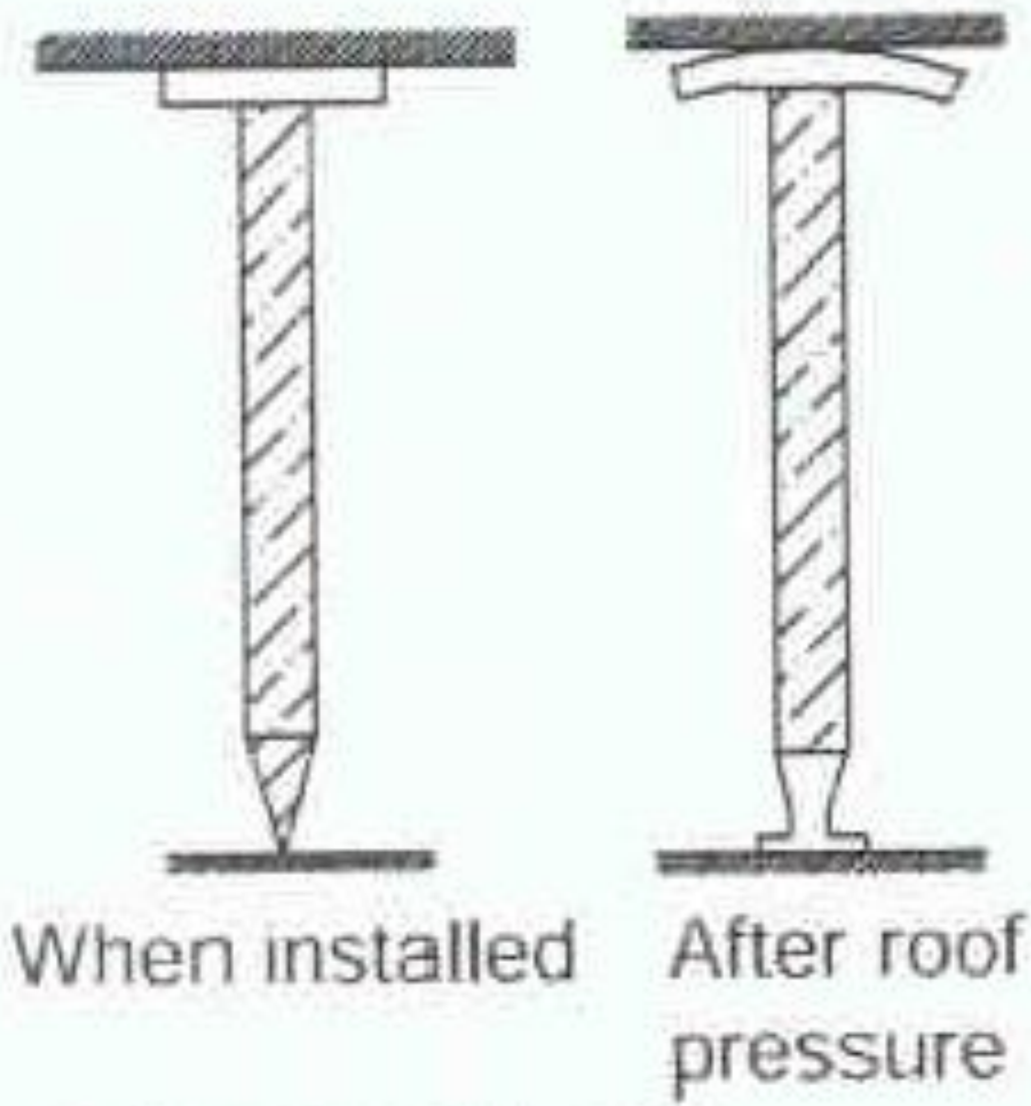
Wooden Support I



(b) Tapered prop

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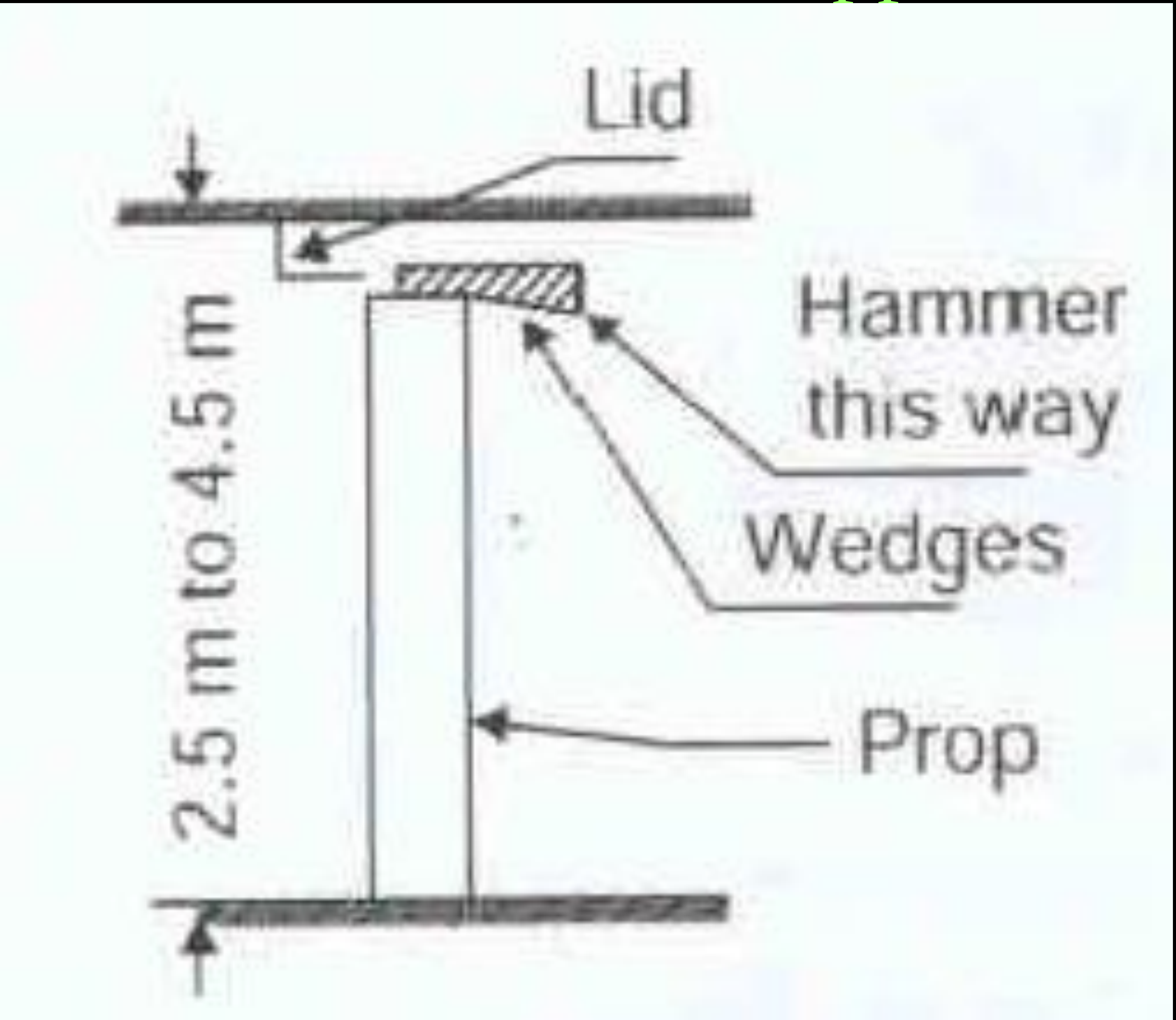
Wooden Support I



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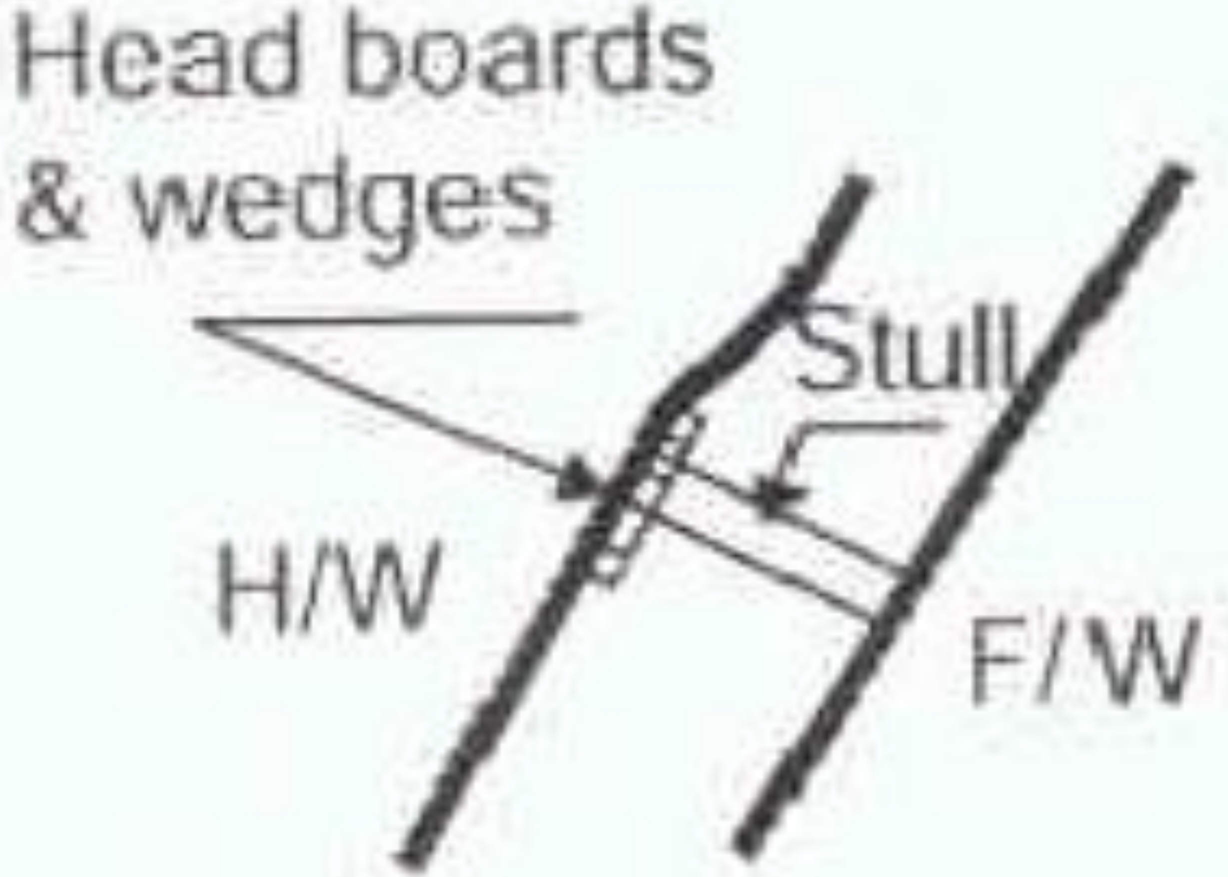
Timber Support I



(c) Round timber stull

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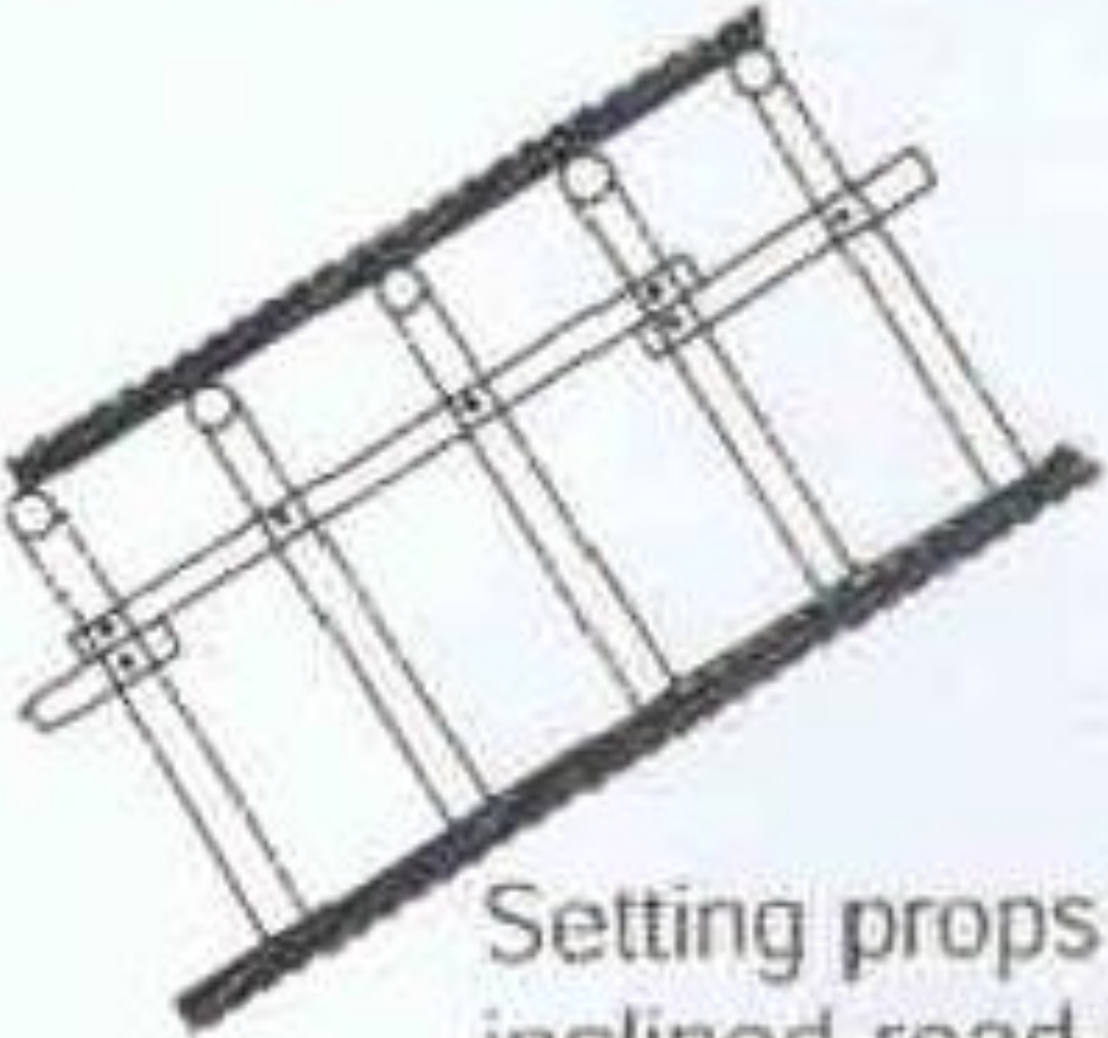
Timber Support I



Setting props in an inclined road-way

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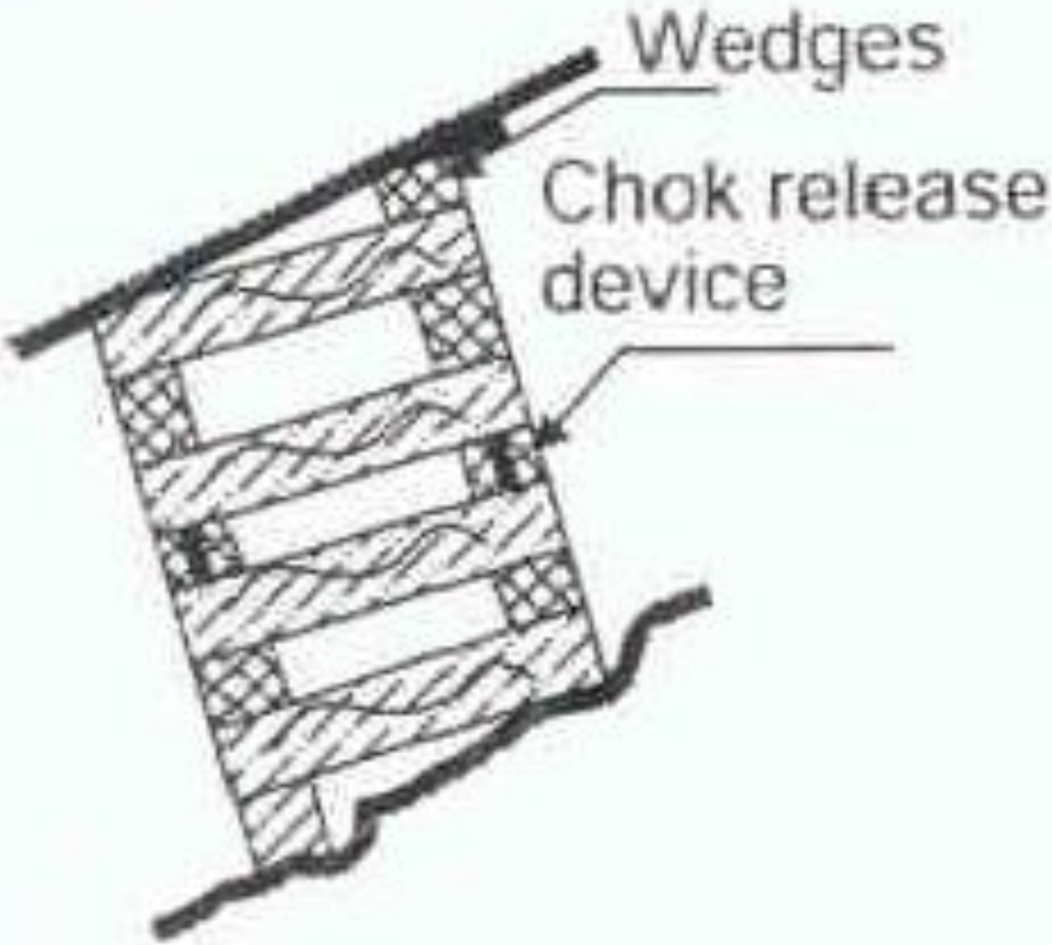
Timber Support I



(d) Chock in an inclined roadway

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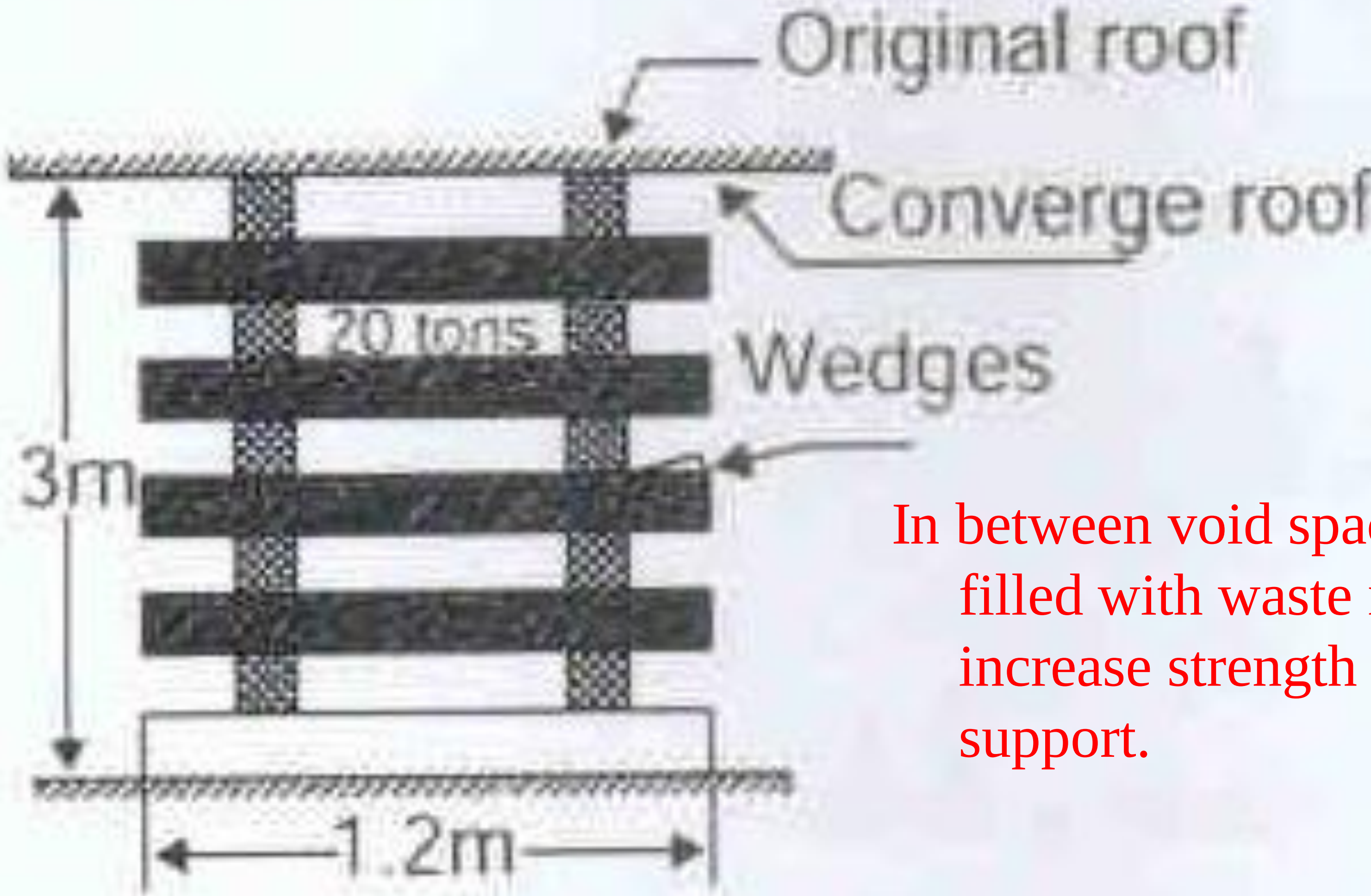
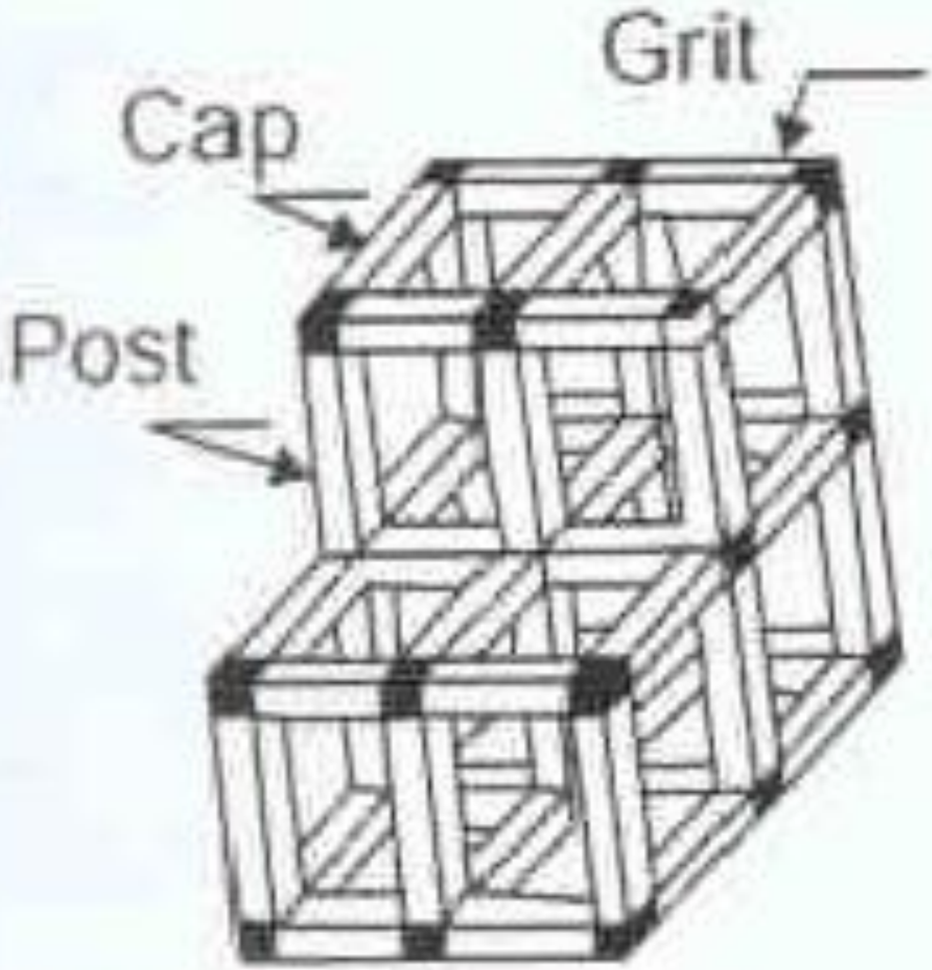
Timber Support I



(e) Wooden chock

In between void space can be filled with waste rock to increase strength of support.

Timber Support I



(f) Assembly of square set modules

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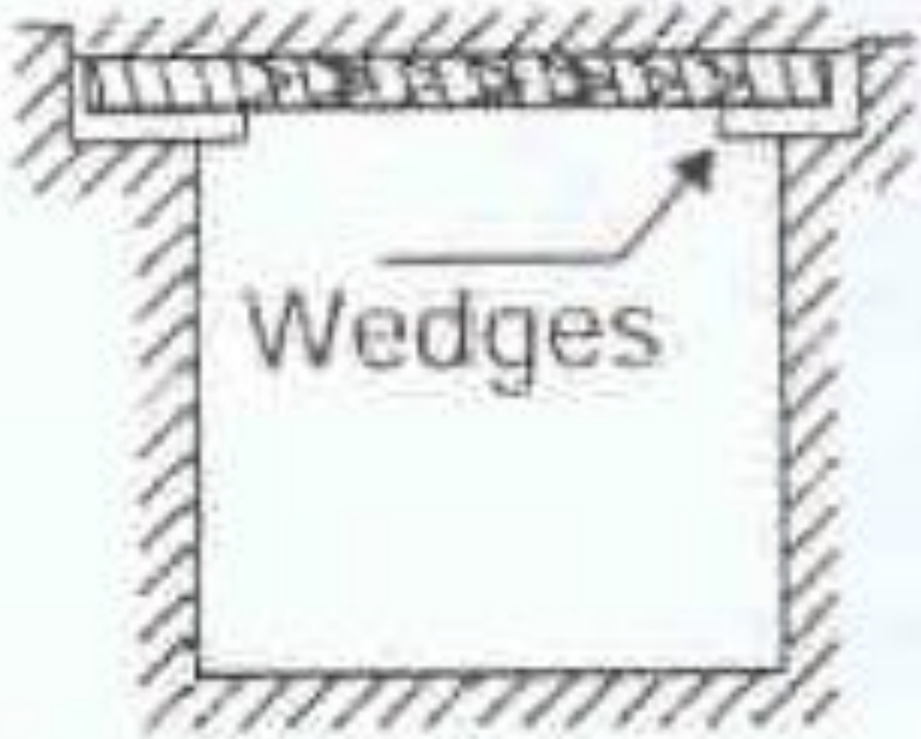
Timber Support I



(g) Wooden bar support

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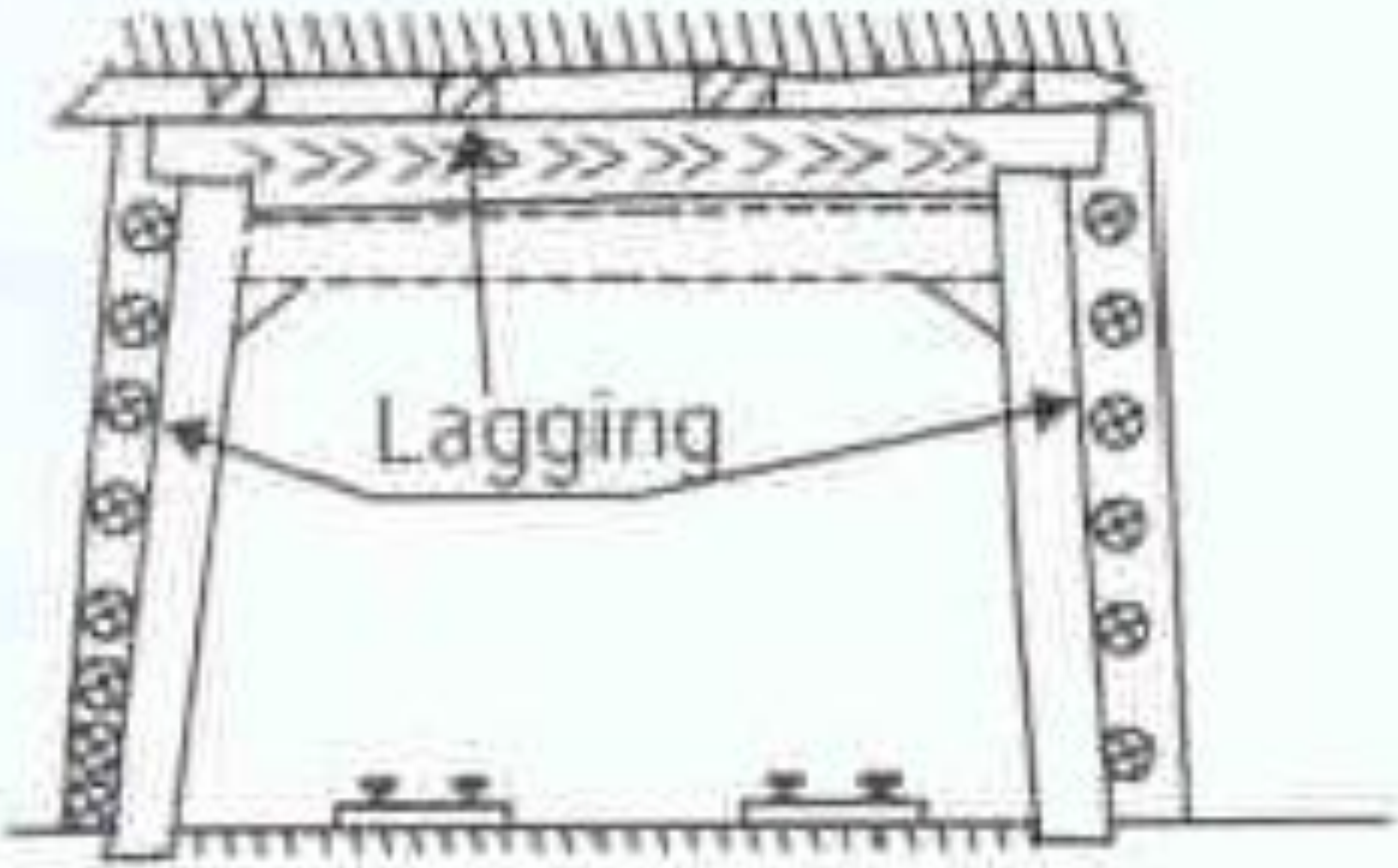
Wooden Support I



(h) Reinforced timber set

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Timber Support II

*Timber gets effected by funguses, that live in humid condition.

*Timber is combatable material & needs due precaution against out bust of fire.

*Freshly cut timber contains above 30% water.

*Before use, timber needs to be dried (seasoned) by natural drying.

*After drying, moisture content in timber gets reduced below 20%.

*Timber has to be made resistant to fungi, bores & insects.

Timber Support III

*Tar & creosotel oil are common organic compound to season timber.

*Salt like zinc chloride, copper sulfate, iron soleplate, lime wash are common inorganic treatment for timber.

*Preservatives are applied by spraying, brushing, cold dipping, hoy & cold open tank treatment, pressure tank treatment,

*At some mines, special treatment is done to make timber fire resistant.

*Timber is made fire

Timber Support IV

*Timber is made fire resistant by applying zinc chloride, ferric chloride, boric acid & ammonium phosphate.

*Timber filled steel pipe, is used as prop (prop stick) South Africa.

*Timber protrudes from each end of steel tube.

*Timber acts as axial load beraing member & steel pipe provides lateral constraints, there by greatly improving yieldability of prop.

*Pipe stitch is a 150-200 mm diameter wooden prop encased in 3-4mm thick steel pipe.

Timber Support V

*Timber is flexible & can accommodate large strain before it breaks.

*Crushing of timber becomes apparent long before it fails.

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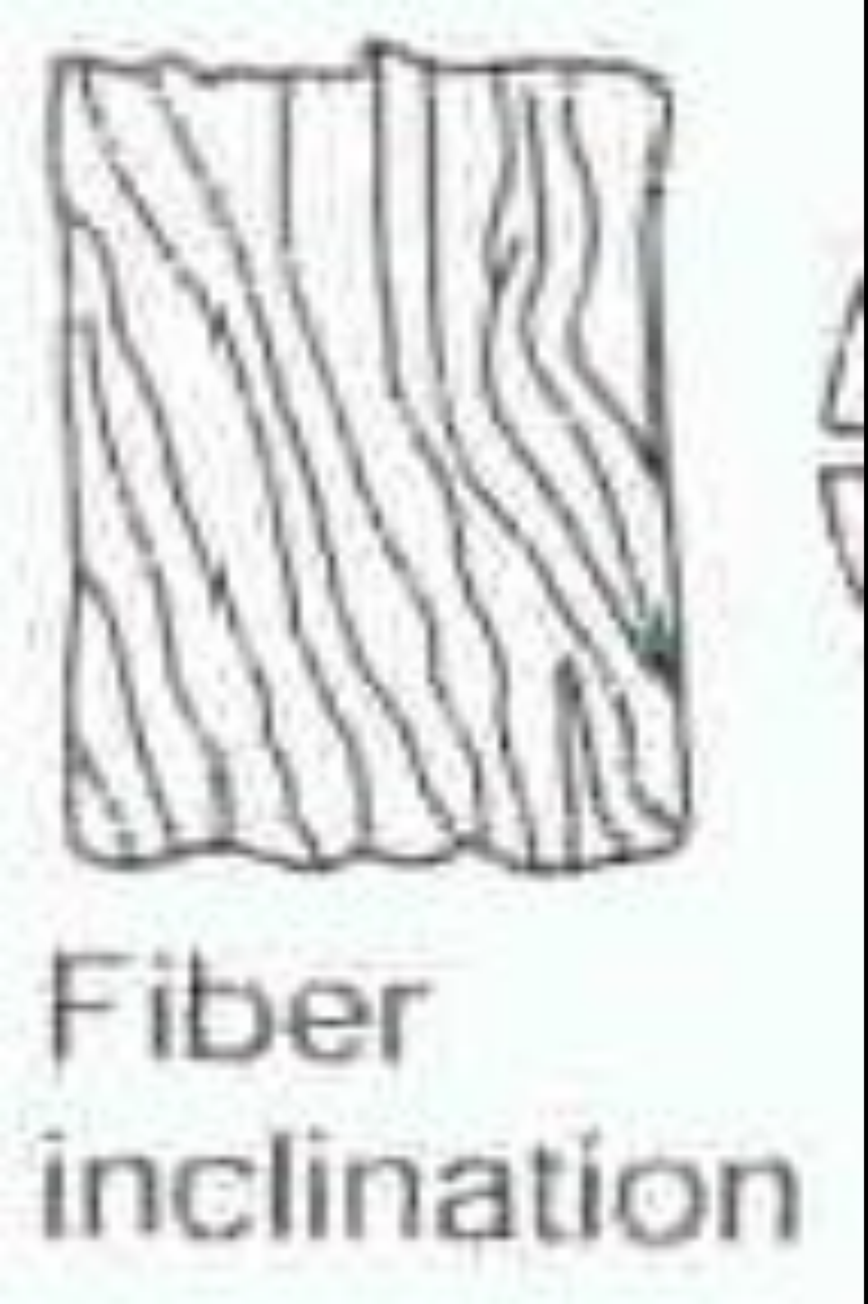
Defects of Timber I

Notch, from where tree branches comes out, also called eye. With time, thia portion becomes hollow.

Defects Of Timber I

Co-centric growth circle indicates uniformity in strength of timber along the cross section. Eccentric growth circle indicate weakness of timber.

Defects of Timber I



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Defects of Timber I

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Defects Of Timber I

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Wooden Support I

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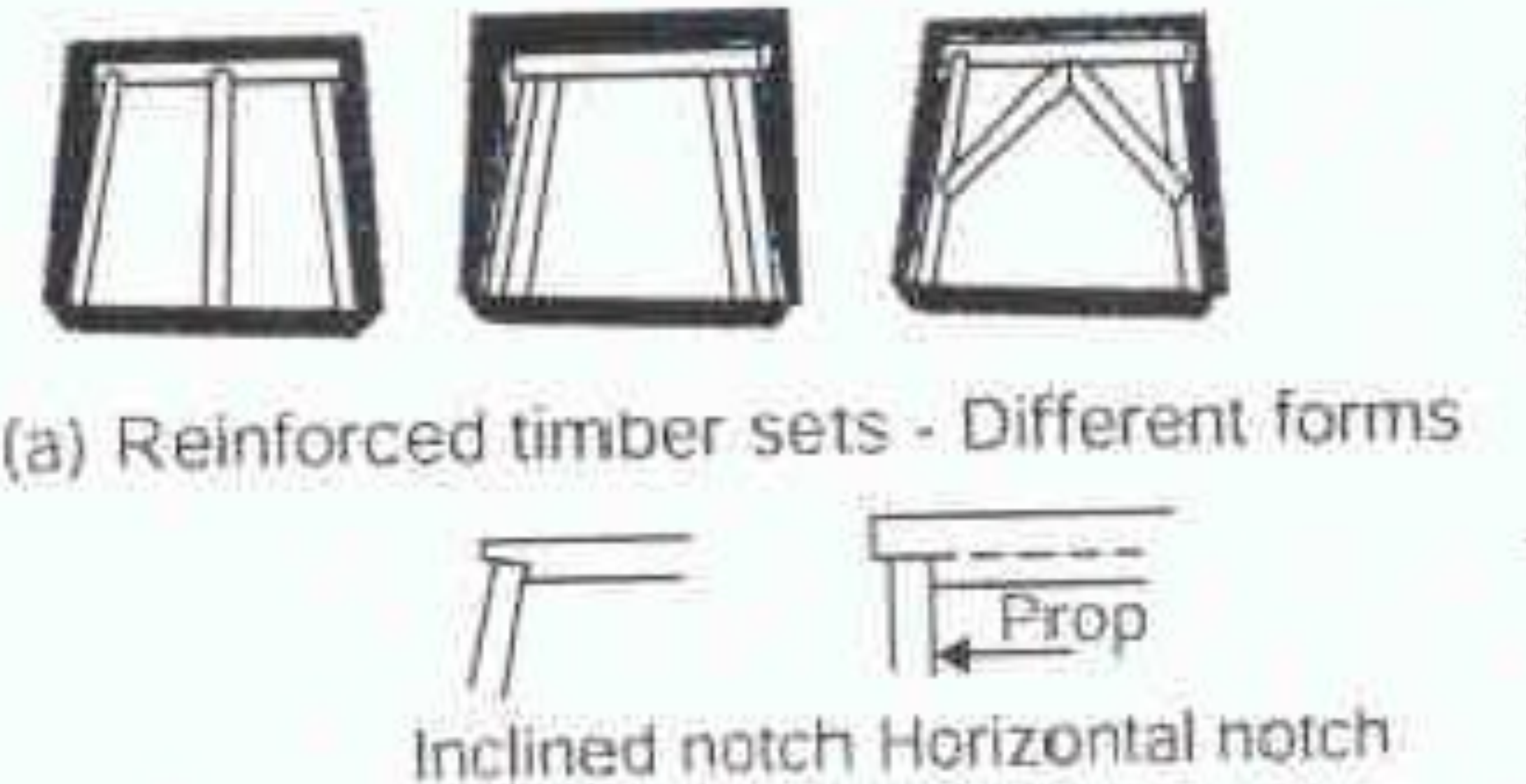
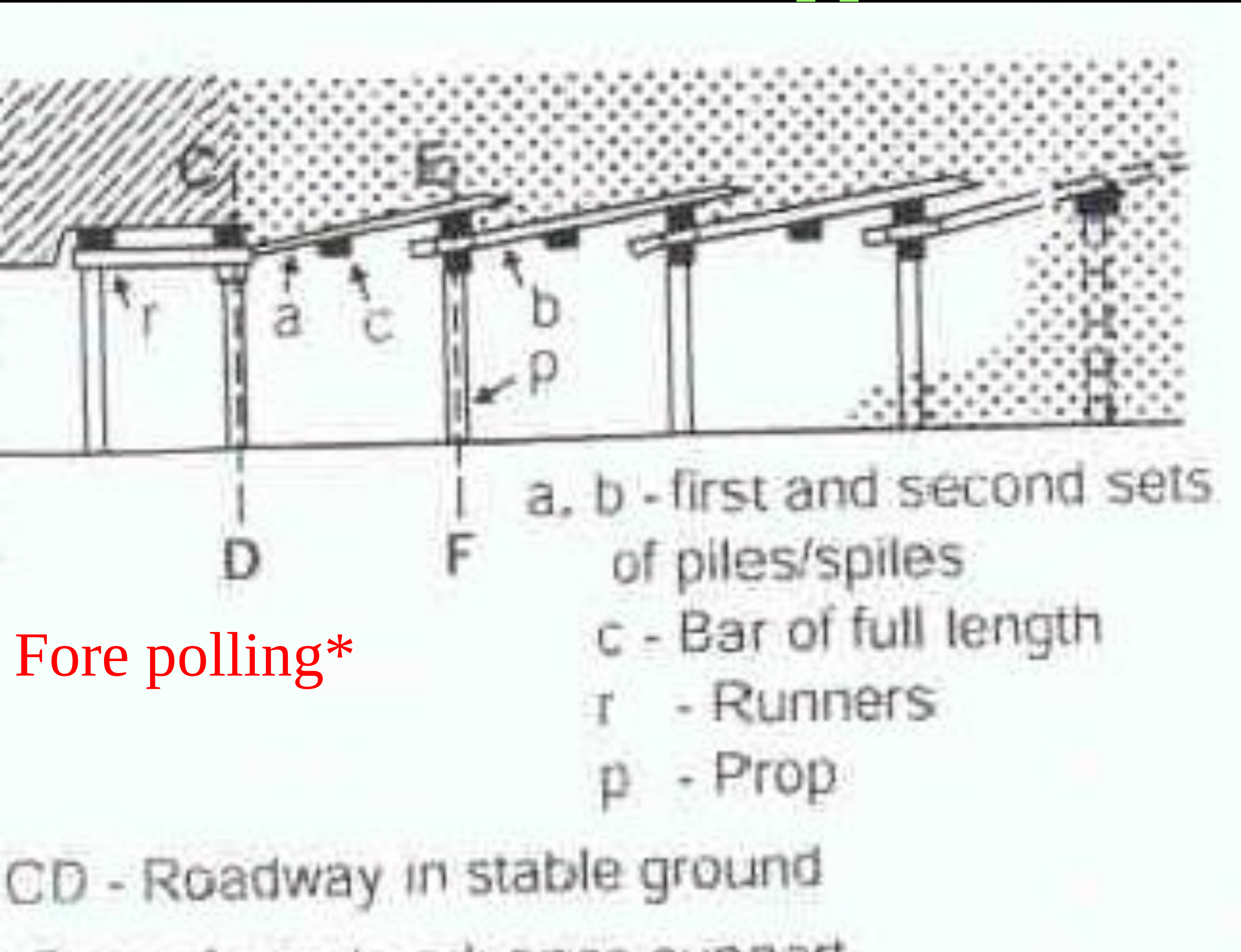


Wooden Support I



Fore polling*

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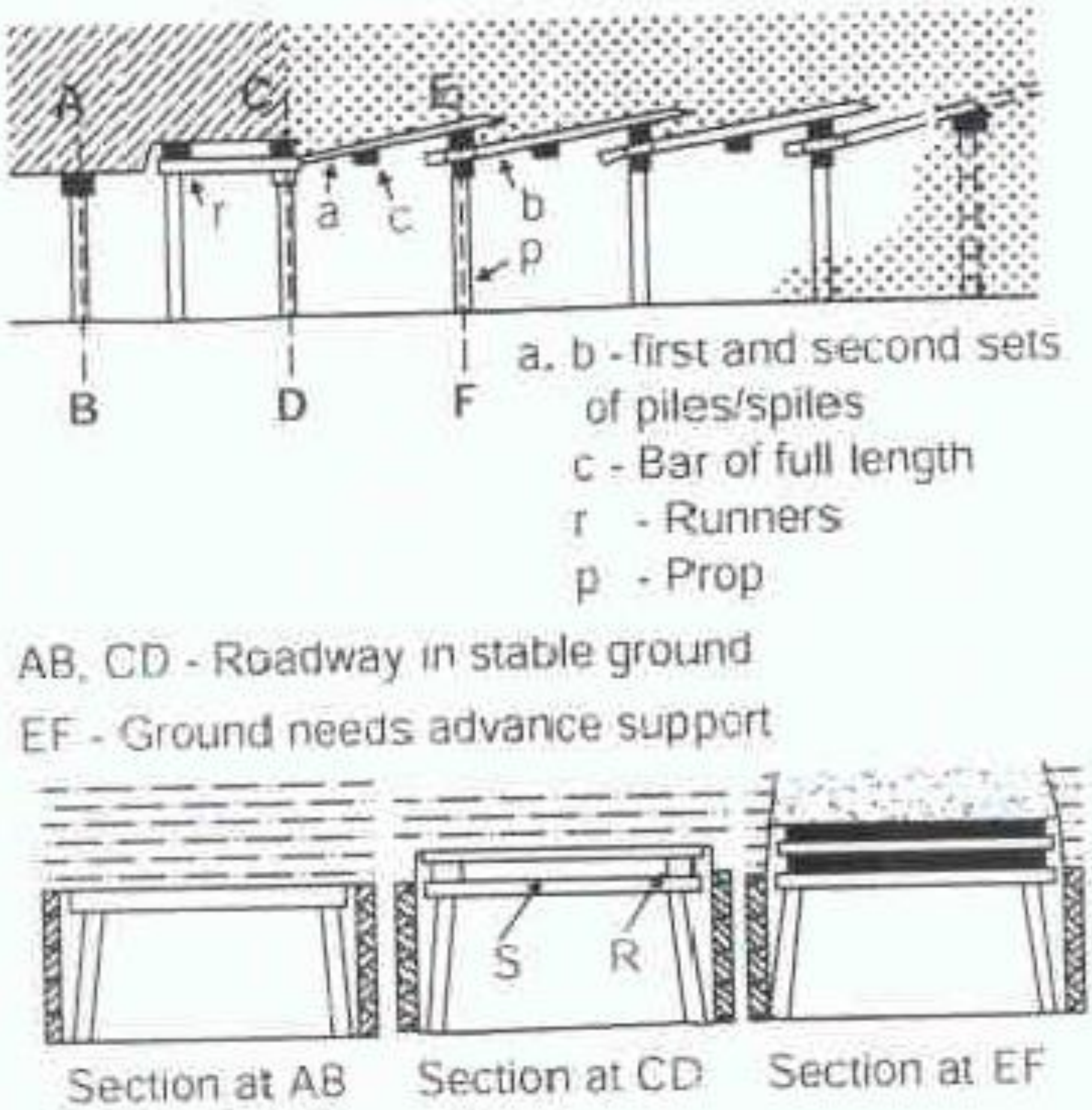
Wooden Support I



(c) Advancing by fore - polling

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Calculation For Timber Support I

Load bearing capacity of prop

$$P = 47.2 - 1.5\ h/d$$

P= load bearing capacity of prop in ton

h= height of prop in mm

d= diameter of prop in mm

Calculation For Timber Support II

Buckling strength of Timber

$B = \pi^2 E/S^2$ for $S>100$, $B=C(1-aS+bS^2)$ for $S<100$

$S=4l/d$

B= buckling strength of timber

E=elasticity modules of timber

S=slenderness ratio

a=0 quality constant for mine timber

b=2 quality constant for mine timber

C= crushing strength of timber

l=length of prop in mm

d=diameter of prop in mm

Calculation For Timber Support III

Bending strength (modules of rapture)

$Bs=M/W$, $M=Pl/4$, $W=bh^2/6$

Bs=bending strength

M=maximum bending movement

W=section modules

P=breaking load

h=height of load in meter

Ll=length of prop

b=2, quality constant for mine timber

Calculation For Timber Support IV

Load on wooden gallery

$$h = AL$$

h=height of prop in mm

A=loading factor, depends on rock formation, under normal condition 0.25 – 0.5, for bad roof with cracks 1-2

Calculation For Timber Support V

Pressure on support in t/m sq

$$Ps=hDr$$

Ps= pressure on support in t/mm sq

Hh= height of prop in mm

Dr=rock density in ton/m sq

Calculation For Timber Support I

Total load produced by parabolic dome per set of support

$$T_L = h\, L_f\, L_a\, D\, S$$

T_L= total load produced by parabolic dome

h=height of load in meter

L_f= loading factor, depends on rock formation under normal condition 0.25 to 0.5, for bad roof with cracks, it may be 1-2.

L_a=span of drive / gallery

D=rock density

S=distance in between two setes of support

Advantages Of Timber Support I

*Timber supports are light in weight.

*Timber can be easily cut, manipulated, transported & out in form of support.

*Timber can be reused.

*Timber gives indication before it fails.

*Timber is best suitable for temporary work.

*Using timber is economical because of low erection cost & multiple use.

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Disadvantages Of Timber Support I

*Timber rots in underground humid climate.

*In dry environment, timber posses' threat of fire in underground, especially in dry hot climate.

*Availability of good suitable timber is becoming poor.

*Timber procurement, handling, use etc. have come under stringent law.

Reference I

*Surface and underground Excavations- Methods, Technique and Equipment. Ratan Raj Tatiya, 2005, page 179 to 185.

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